EMC TECHNICAL FILE

Product Name	:	AUTOMATIC BRASS BUSBAR MACHINE & AUTOMATIC SCREW MACHINE
Model Name	:	MODEL YMACT-666 & MODEL YMSM-888

Prepared for:

ZHEJIANG YOMIN ELECTRIC CO.,LTD.
QIAOQIAN INDUSTRY AREA, LIUSHI, YUEQING,ZHEJIANG CHINA.

Prepared by:

Shanghai Global Testing Services Co., Ltd.
Floor 3rd, Building D-1, No. 128, Shenfu Road, Minhang District,
Shanghai, China.

Report Number	:	TEZJ25062572397
Date of Report	:	July 11, 2025
Date of Test	:	July 01, 2025 to July 11, 2025

Notes：
The review results only relate to these samples which have been reviewed.
Partly using this file will not be admitted unless been allowed by GTS.
GTS is only responsible for the complete file with the fileed stamp of GTS.

Applicant:	ZHEJIANG YOMIN ELECTRIC CO.,LTD. QIAOQIAN INDUSTRY AREA, LIUSHI, YUEQING,ZHEJIANG CHINA.
Manufacturer:	ZHEJIANG YOMIN ELECTRIC CO.,LTD. QIAOQIAN INDUSTRY AREA, LIUSHI, YUEQING,ZHEJIANG CHINA.
Product Name:	AUTOMATIC BRASS BUSBAR MACHINE & AUTOMATIC SCREW MACHINE
Brand Name:	/
Model Name:	MODEL YMACT-666 & MODEL YMSM-888
Main Model:	YMACT-666
Serial Number:	/
Rating:	440V
Date of Receipt:	July 01, 2025
Date of Test:	July 01, 2025 to July 11, 2025
Test Standard:	EN IEC 61000-6-1:2019, EN IEC 61000-6-3:2021, EN IEC 61000-3-2:2019+A2:2024, EN 61000-3-3:2013+A2:2021+AC:2022
Test Result:	PASS

Prepared by : Anson Qin

Approved by : Stephen Yang

GLOBAL TESTING SERVICES CO.,LTD. GTS CERTIFICATION

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1. GENERAL INFORMATION

1.1 Description of EUT

Product Name:	AUTOMATIC BRASS BUSBAR MACHINE & AUTOMATIC SCREW MACHINE
Model Name:	YMACT-666
Serial Number:	N/A
Power Supply	/
Applicant:	ZHEJIANG YOMIN ELECTRIC CO.,LTD. QIAOQIAN INDUSTRY AREA, LIUSHI, YUEQING,ZHEJIANG CHINA.
Manufacturer:	ZHEJIANG YOMIN ELECTRIC CO.,LTD. QIAOQIAN INDUSTRY AREA, LIUSHI, YUEQING,ZHEJIANG CHINA.

1.2 Description of Test Facility

Site Description:	Shanghai Global Testing Services Co., Ltd.
Name of Firm:	Shanghai Global Testing Services Co., Ltd
Site Location:	Floor 3rd, Building D-1, No. 128, Shenfu Road, Minhang District, Shanghai, China.
The site and apparatus are constructed in conformance with the requirements of ANSI C63.4, CISPR 16-1-1 and other equivalent standards.	

1.3 Measurement Uncertainty

Conducted Emission Expanded Uncertainty:	U = 1.76 dB
Radiated Emission Expanded Uncertainty:	U = 3.02 dB

2. TECHNICAL SUMMARY

2.1 SUMMARY OF STANDARDS AND TEST RESULTS

The EUT have been tested according to the applicable standards as referenced below:

EN IEC 61000-6-3:2021			
Test Item	**Test Standard**	**Limits**	**Results**
Conducted Disturbance at low voltage AC mains ports	CISPR 16-1-1:2015,CISPR 16-1-2:2014,CISPR 16-2-1:2014	See 4.3	P
Radiated Disturbance	CISPR 16-2-3:2016	See 7.3	P
Harmonic Current Emissions	EN IEC 61000-3-2:2019+A2:2024	/	P
Voltage Fluctuations and Flicker	EN 61000-3-3:2013+A2:2021+AC:2022	/	P

EN IEC 61000-6-1:2019			
Test Item	**Basic Standard**	**Performance Criteria**	**Results**
Electrostatic discharge Immunity	IEC 61000-4-2:2008	B	P
RF Electromagnetic Field Immunity	IEC 61000-4-3:2020	A	P
Electrical Fast Transient/Burst Immunity	IEC 61000-4-4:2012	B	P
Surge Immunity	IEC 61000-4-5:2014+A1:2017	B	P
Conducted Disturbances Immunity	IEC 61000-4-6:2013+Cor 1:2015	A	P
Power-frequency Magnetic Field Immunity	IEC 61000-4-8:2009	A	P

Note: P means pass, F means failure, N/A means not applicable

2.2 Description of Performance Criteria

The variety and the diversity of the apparatus within the scope of this standard make it difficult to define precise criteria for the evaluation of the immunity test results.
If, as result of the application of the tests defined in this standard, the apparatus becomes dangerous or unsafe, the apparatus shall be deemed to have failed the test.
A functional description and a definition of performance criteria, during or as a consequence of the EMC testing, shall be provided by the manufacturer and noted in the test report, based on the following criteria:

2.2.1 Performance criterion A
The apparatus shall continue to operate as intended during the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and from what the user may reasonably expect from the apparatus if used as intended.

2.2.2 Performance criterion B
The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is allowed however. No change of actual operating state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and from what the user may reasonably expect from the apparatus if used as intended.

2.2.3 Performance criterion C
Temporary loss of function is allowed, provided the function is self-recoverable or can be restored by the operation of the controls, or by any operation specified in the instructions for use.

3. TEST EQUIPMENT LIST

Conducted Disturbance at DC power ports				
Equipment	**Manufacturer**	**Model**	**Serial No.**	**Next Cal.**
Shielding Room	CHENGYU	5m×4m×3m	CR	Dec 19, 2025
EMI Test Receiver	R&S	ESCI7	100787	Dec 19, 2025
Artificial Mains Network	TESEQ	NNB 51	33285	Dec 19, 2025

Radiated Disturbance Test				
Equipment	**Manufacturer**	**Model**	**Serial No.**	**Next Cal.**
3m Semi-anechoic Chamber	CHENGYU	9.2×6.25×6.15m	SAR	Dec 19, 2025
EMI Test Receiver	R&S	ESCI7	100787	Dec 19, 2025
EMC Shielding room	Changzhou FeiTe	8 x 5 x 3 mm	Nil	Dec 19, 2025
Broadband Log Antenna	Schwarzbeck	VULB 9163	9163-561	Dec 19, 2025

Electrostatic Discharge Immunity Test				
Equipment	**Manufacturer**	**Model**	**Serial No.**	**Next Cal.**
ESD Generator	SCHAFFNER	NSG 438	849	Dec 19, 2025

RF Electromagnetic Field Immunity				
Equipment	**Manufacturer**	**Model**	**Serial No.**	**Next Cal.**
Radiated Immunity Test System	TESEQ	ITS 6006	37546	Dec 19, 2025
Power Meter	TESEQ	PMR 6006	73819	Dec 19, 2025
Power Amplifier	MILMEGA	AS1860-50	1066592	Dec 19, 2025
Log Periodic Antenna	Schwarzbeck	STLP 9128 D	9128 D 048	Dec 19, 2025
Field Probe	ETS-Lindgren	HI-6105	00161798	Dec 19, 2025

Electrical Fast Transient/SURGE Immunity Test				
Equipment	**Manufacturer**	**Model**	**Serial No.**	**Next Cal.**
EFT/SURGE Generator	TESEQ	NSG 3060	1468	Dec 19, 2025
Single Phase Coupling/decoupling Network	TESEQ	CDN 3061	1404	Dec 19, 2025
Capacitive clamp	TESEQ	CDN 3425	1736	Dec 19, 2025

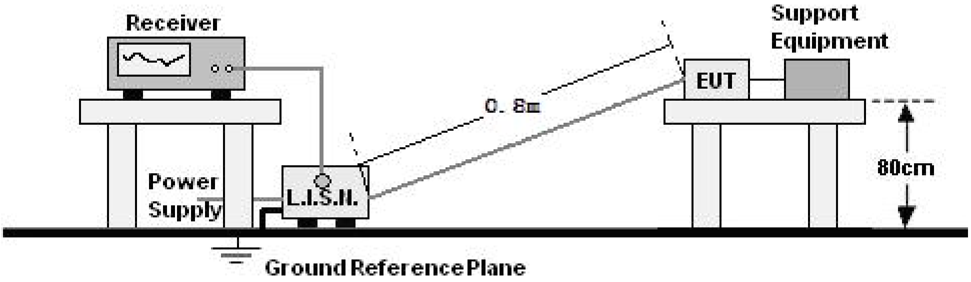
Conducted Disturbances Immunity Test				
Equipment	**Manufacturer**	**Model**	**Serial No.**	**Next Cal.**
Conducted Immunity Test System	TESEQ	NSG 4070	25795	Dec 19, 2025
Coupling/Decoupling Network	TESEQ	CDN M116S	35371	Dec 19, 2025
EM-clamp	TESEQ	KEMZ 801	29530	Dec 19, 2025

Power-frequency Magnetic Field Immunity Test				
Equipment	**Manufacturer**	**Model**	**Serial No.**	**Next Cal.**
P-f Magnetic Field Loop	FCC	F-1000-4-8/9/10-1M	13	Dec 19, 2025
Power Magnetic Field Generator	SANKI	SKS-0805	/	Dec 19, 2025

The measuring equipment utilized to perform the tests documented in this report has been calibrated once a year or in accordance with the manufacturer's recommendations, and has been calibrated by accredited calibration laboratories.

4. CONDUCTED DISTURBANCE AT LOW VOLTAGE AC MAINS PORTS

4.1 DIAGRAM OF TEST SETUP



4.2 APPLICABLE STANDARD

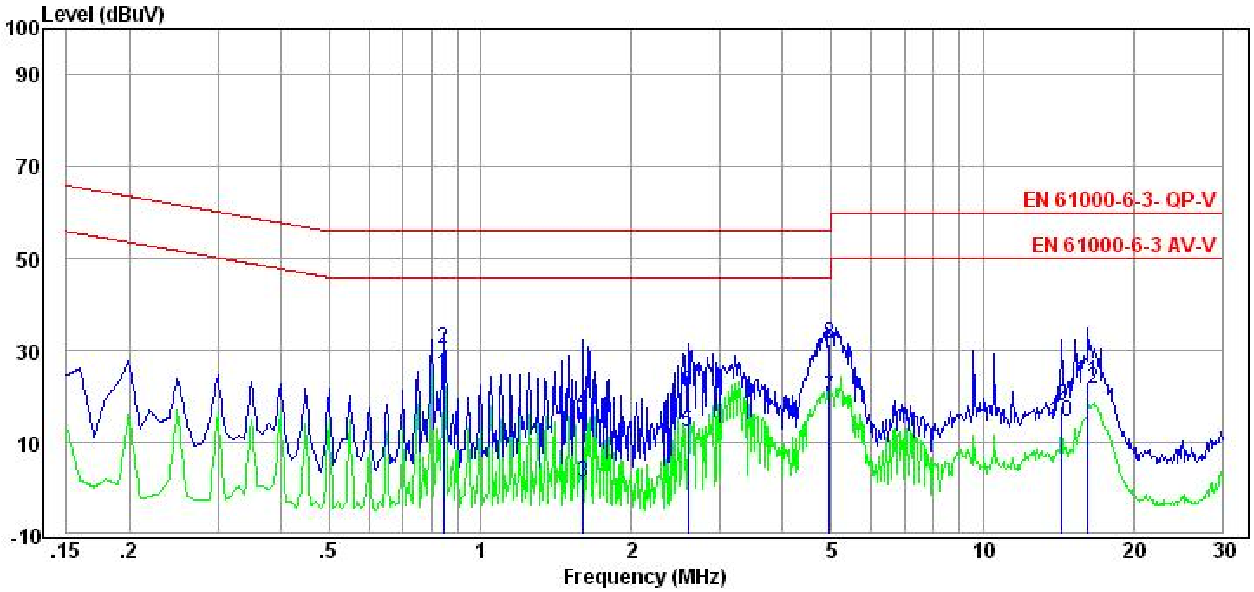
EN IEC 61000-6-3:2021(CISPR 16-1-1:2015, CISPR 16-1-2:2014, CISPR 16-2-1:2014)

4.3 LIMITS FOR CONDUCTED DISTURBANCE

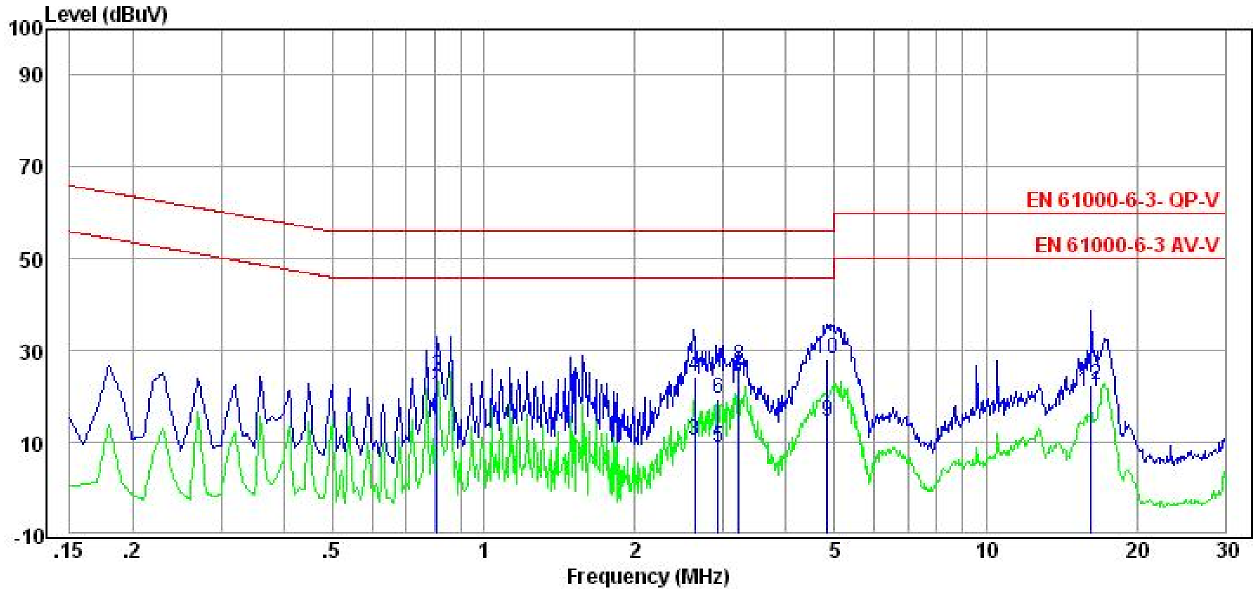
Frequency Range (MHz)	Limits dB(μV)	
	Quasi-peak	Average
0.15 ～ 0.5	66-56	56-46
0.5 ～ 5	56	46
5 ～ 30	60	50
NOTE – The lower limit shall apply at the transition frequencies.		

4.4 TEST RESULT

TEMPERATURE:	22℃	HUMIDITY:	43%
TEST MODEL:	Operating	Voltage supply:	440V



Freq MHz	Reading dBuV	C.F dB	Result dBuV	Limit dBuV	Margin dB	Detector
0.85	15.03	9.70	24.73	46.00	21.27	Average
0.85	20.74	9.70	30.44	56.00	25.56	QP
1.61	-8.32	9.74	1.42	46.00	44.58	Average
1.61	6.94	9.74	16.68	56.00	39.32	QP
2.60	2.30	9.87	12.17	46.00	33.83	Average
2.60	11.26	9.87	21.13	56.00	34.87	QP
4.95	10.01	9.92	19.93	46.00	26.07	Average
4.95	21.62	9.92	31.54	56.00	24.46	QP
14.37	7.86	10.03	17.89	50.00	32.11	Average
14.37	4.48	10.03	14.51	60.00	45.49	QP
16.21	13.11	10.12	23.23	50.00	26.77	Average
16.21	11.67	10.12	21.79	60.00	38.21	QP



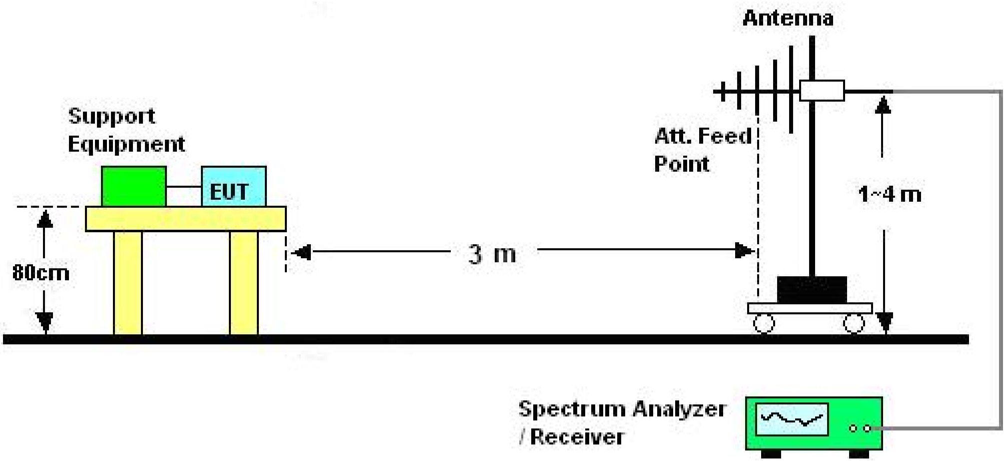
Freq MHz	Reading dBuV	C.F dB	Result dBuV	Limit dBuV	Margin dB	Detector
0.81	10.50	9.69	20.19	46.00	25.81	Average
0.81	14.81	9.69	24.50	56.00	31.50	QP
2.63	0.51	9.89	10.40	46.00	35.60	Average
2.63	14.41	9.89	24.30	56.00	31.70	QP
2.93	-0.88	9.91	9.03	46.00	36.97	Average
2.93	9.74	9.91	19.65	56.00	36.35	QP
3.23	14.34	9.89	24.23	46.00	21.77	Average
3.23	16.65	9.89	26.54	56.00	29.46	QP
4.83	4.58	9.91	14.49	46.00	31.51	Average
4.83	18.15	9.91	28.06	56.00	27.94	QP
16.24	11.00	10.14	21.14	50.00	28.86	Average
16.24	12.48	10.14	22.62	60.00	37.38	QP

4.5 TEST CONCLUSION

PASS

5. RADIATED DISTURBANCE TEST

5.1 DIAGRAM OF TEST SETUP



5.2 APPLICABLE STANDARD

EN IEC 61000-6-3:2021 (CISPR 16-2-3:2016)

5.3 LIMITS FOR RADIATED DISTURBANC

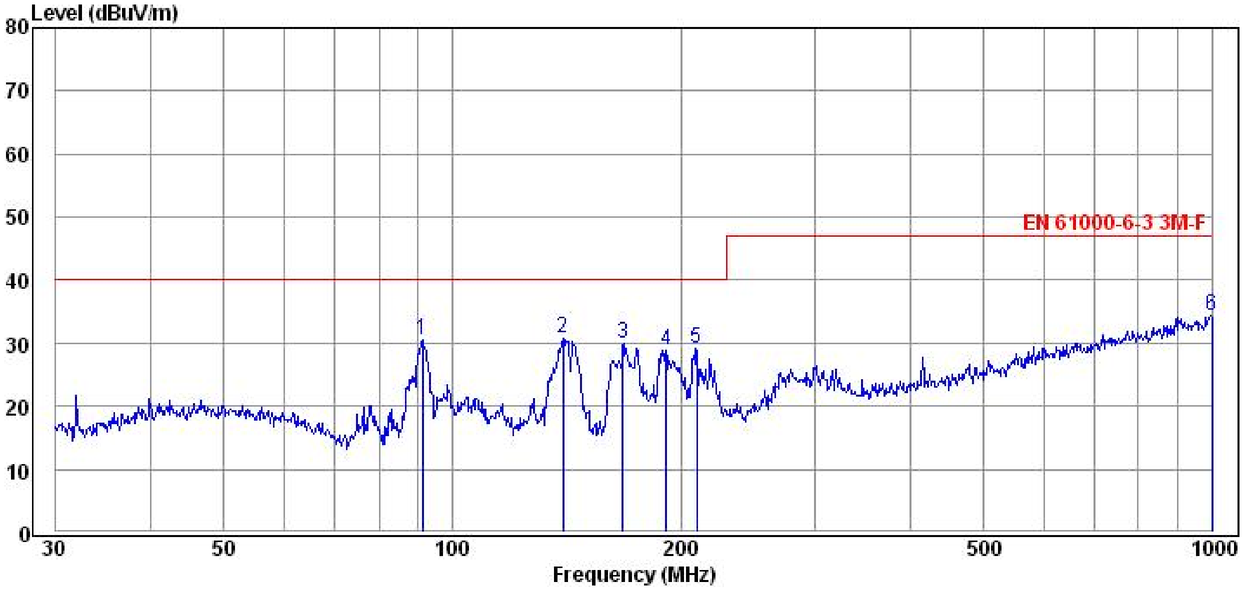
Below 1GHz

Frequency (MHz)	Distance (m)	Field Strength Limits dB(V/m)	Converted Field Strength Limits By 3 Meters Measuring Distance dB(V/m)
30 ~ 230	10	30	40
230 ~ 1000	10	37	47
NOTE 1 - The lower limit shall apply at the transition frequency. NOTE 2 – Additional provisions may be required for cases where interference occurs.			

5.4 TEST RESULT

TEMPERATURE:	22℃	HUMIDITY:	43%
TEST MODEL:	Operating	Voltage supply:	440V

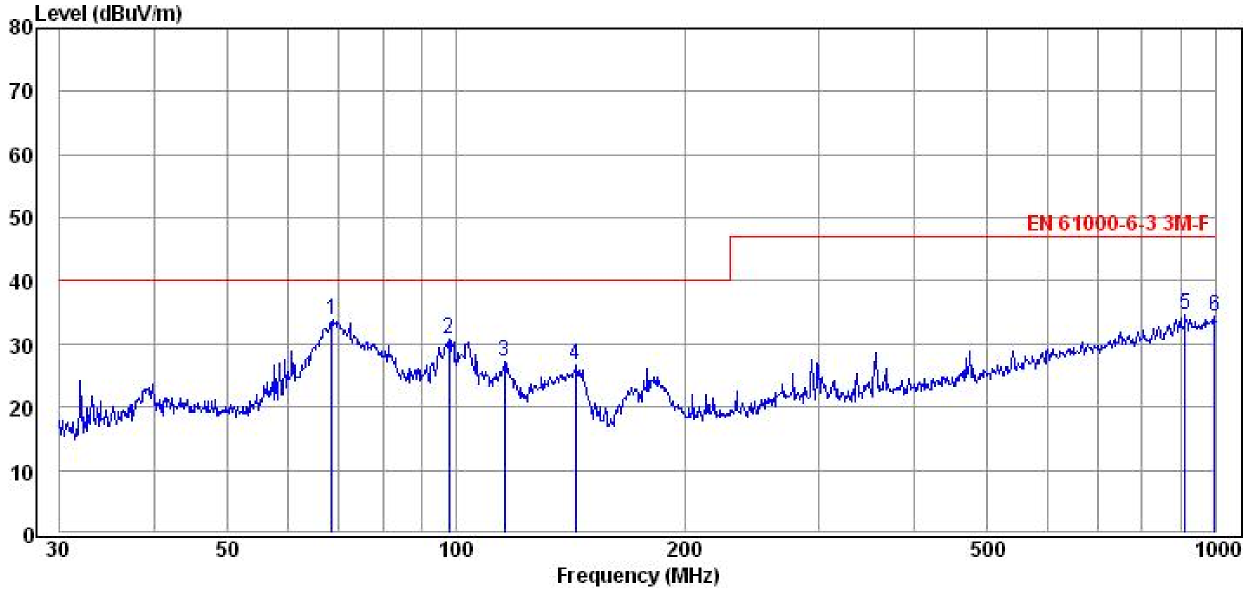
HORIZONTAL:



Freq MHz	Reading dBuV	C.F dB	Result dBuV	Limit dBuV	Margin dB	Detector
91.31	17.73	12.84	30.57	40.00	9.43	QP
139.76	19.85	10.91	30.76	40.00	9.24	QP
167.57	18.14	11.78	29.92	40.00	10.08	QP
191.32	14.98	13.86	28.84	40.00	11.16	QP
209.49	14.97	14.17	29.14	40.00	10.86	QP
1000.00	5.40	29.03	34.43	47.00	12.57	QP

Remarks: C.F (Correction Factor) = Antenna factor + Cable loss - Preamp gain

VERTICAL：



Freq MHz	Reading dBuV	C.F dB	Result dBuV	Limit dBuV	Margin dB	Detector
68.46	21.87	11.92	33.79	40.00	6.21	QP
97.80	16.65	14.16	30.81	40.00	9.19	QP
115.66	13.49	13.63	27.12	40.00	12.88	QP
143.66	15.74	10.88	26.62	40.00	13.38	QP
913.10	6.67	28.07	34.74	47.00	12.26	QP
996.50	5.34	28.98	34.32	47.00	12.68	QP

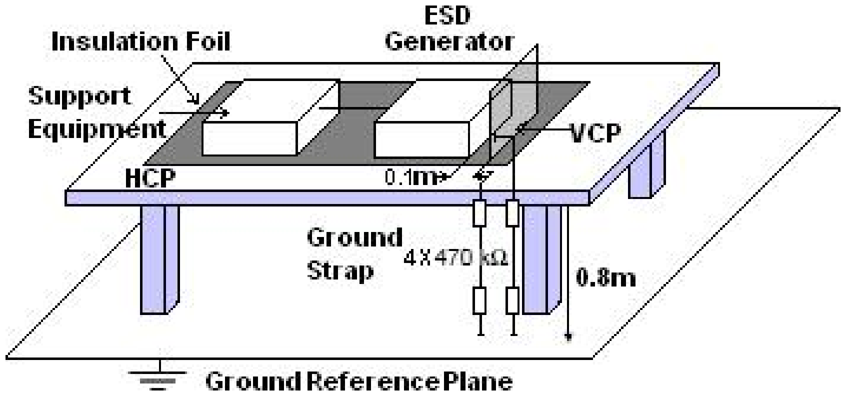
Remarks: C.F (Correction Factor) = Antenna factor + Cable loss - Preamp gain

5.5 TEST CONCLUSION

PASS

6. ELECTROSTATIC DISCHARGE IMMUNITY TEST

6.1 DIAGRAM OF TEST SETUP



6.2 APPLICABLE STANDARD

IEC 61000-4-2:2008, Contact Discharge:±4kV;

Air Discharge: ±8kV

6.3 SEVERITY LEVELS AND PERFORMANCE CRITERION

6.3.1 Severity levels

Level	Test Voltage	
	Contact Discharge (kV)	Air Discharge (kV)
1.	2	2
2.	4	4
3.	6	8
4.	8	15
X	Special	Special

6.3.2 Performance criterion: B

6.4 TEST RESULT

TEMPERATURE:	22℃	HUMIDITY:	43%
TEST MODEL:	Operating	Voltage supply:	440V

Air Discharge Voltage: ±2kV, ±4kV, ±8kV
Contact Discharge Voltage: ±2kV, ±4kV
Contact Discharge: For each point positive 10 times and negative 10 times discharge
Air Discharge: For each point positive 10 times and negative 10 times discharge

Location	Point	Kind	Result
Around the EUT	4	C (VCP)	A
Around the EUT	4	C (HCP)	A
Metal part of EUT and screws	20	C	B
Gap and Button	/	A	B

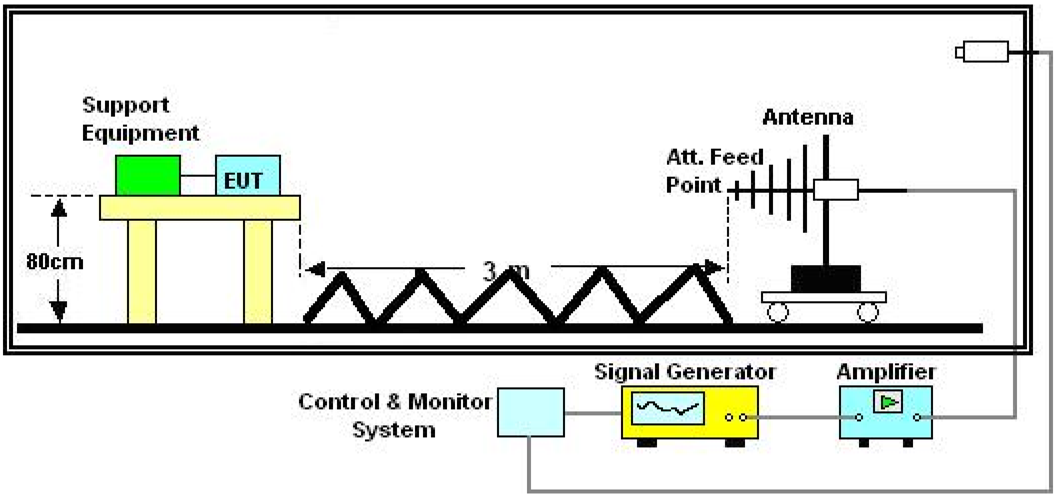
NOTE 1 – C (Contact Discharge), A(Air Discharge);
NOTE 2 – HCP (Horizontal Coupling Plane), VCP (Vertical Coupling Plane).

6.5 TEST CONCLUSION

PASS

7. RF ELECTROMAGNETIC FIELD IMMUNITY TEST

7.1 Diagram of Test Setup



7.2 Applicable Standard

IEC 61000-4-3:2020,
Frequency Range:

80 - 1000 MHz, Field Strength: 10 V/m, Unmodulation:, 80% AM 1kHz;

1400 - 2000 MHz, Field Strength: 10V/m, Unmodulation, 80% AM 1kHz; 2000 - 2700 MHz, Field Strength: 10 V/m, Unmodulation, 80% AM 1kHz

7.3 Severity Levels and Performance Criterion

7.3.1 Severity levels

Level	Field Strength V/m
1	1
2	3
3	10
X	Special

7.3.2 Performance criterion: A

7.4 Test Result

TEMPERATURE:	22℃	HUMIDITY:	43%
TEST MODEL:	Operating	Voltage supply:	440V

<table>
<tr><td colspan="2">Frequency Range</td><td colspan="2">80 MHz to 1000 MHz</td><td colspan="2">1400 MHz to 2000 MHz</td><td colspan="2">2000 MHz to 2700 MHz</td></tr>
<tr><td colspan="2">Modulation</td><td colspan="2">80% AM 1 kHz</td><td colspan="2">80% AM 1 kHz</td><td colspan="2">80% AM 1 kHz</td></tr>
<tr><td colspan="2">Steps</td><td colspan="2">1 %</td><td colspan="2">1 %</td><td colspan="2">1 %</td></tr>
<tr><td colspan="2">Dwell Time</td><td colspan="2">3 s</td><td colspan="2">3 s</td><td colspan="2">3 s</td></tr>
<tr><td colspan="2">Antenna Polarization</td><td colspan="2">80 MHz to 1000 MHz</td><td colspan="2">1400 MHz to 2000 MHz</td><td colspan="2">2000 MHz to 2700 MHz</td></tr>
<tr><td colspan="2">Field Strength</td><td colspan="2">10V/m</td><td colspan="2">10V/m</td><td colspan="2">10V/m</td></tr>
<tr><td colspan="2">Antenna Polarization</td><td>Horizontal</td><td>Vertical</td><td>Horizontal</td><td>Vertical</td><td>Horizontal</td><td>Vertical</td></tr>
<tr><td rowspan="6">EUT Position</td><td>Front</td><td>A</td><td>A</td><td>A</td><td>A</td><td>A</td><td>A</td></tr>
<tr><td>Rear</td><td>A</td><td>A</td><td>A</td><td>A</td><td>A</td><td>A</td></tr>
<tr><td>Right</td><td>A</td><td>A</td><td>A</td><td>A</td><td>A</td><td>A</td></tr>
<tr><td>Left</td><td>A</td><td>A</td><td>A</td><td>A</td><td>A</td><td>A</td></tr>
<tr><td>Floor</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td></tr>
<tr><td>Top</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td></tr>
<tr><td colspan="8">NOTE – “--” means the item is no applicable.</td></tr>
</table>

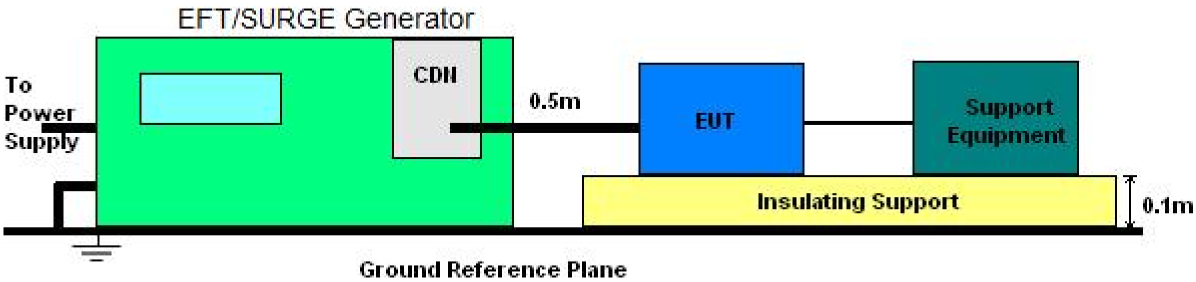
7.5 TEST CONCLUSION

PASS

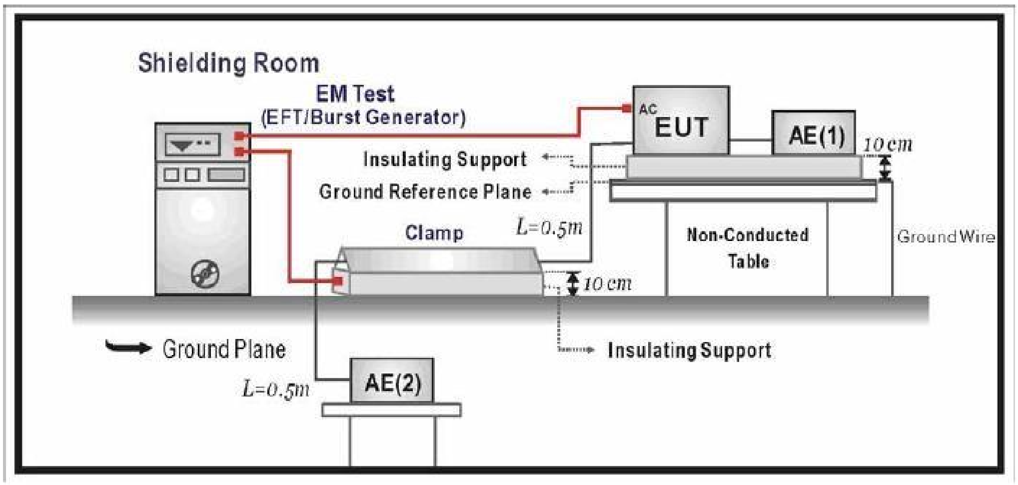
8. ELECTRICAL FAST TRANSIENT/BURST IMMUNITY TEST

8.1 DIAGRAM OF TEST SETUP

FOR POWER PORT：



FOR SIGNAL PORT:



8.2 APPLICABLE STANDARD

IEC 61000-4-4:2012, Signal ports and AC Power ports：±2 kV, 5/50ns, 5kHz

8.3 EVERITY LEVELS AND PERFORMANCE CRITERION

8.3.1 SEVERITY LEVELS

Open circuit output test voltage and repetition rate of the impulses				
Level	On power port, PE		On I/O (input/output) signal, data and control ports	
	Voltage peak kV	Repetition rate kHz	Voltage peak kV	Repetition rate kHz
1.	0.5	5 or 100	0.25	5 or 100
2.	1	5 or 100	0.5	5 or 100
3.	2	5 or 100	1	5 or 100
4.	4	5 or 100	2	5 or 100
Xa	Special	Special	Special	Special

Note 1: Use of 5kHz repetition rates is traditional; however, 100kHz is closer to reality. Product committees should determine which frequencies are relevant for specific products or product types. Note 2: With some products, there may be no clear distinction between power ports and I/O ports, in which case it is up to product committees to make this determination for test purposes.
“Xa” is an open level. The level has to be specified in the dedicated equipment specification.

8.3.2 PERFORMANCE CRITERION: B

8.4 TEST RESULTS

TEMPERATURE:	22℃	HUMIDITY:	43%
TEST MODEL:	Operating	Voltage supply:	440V

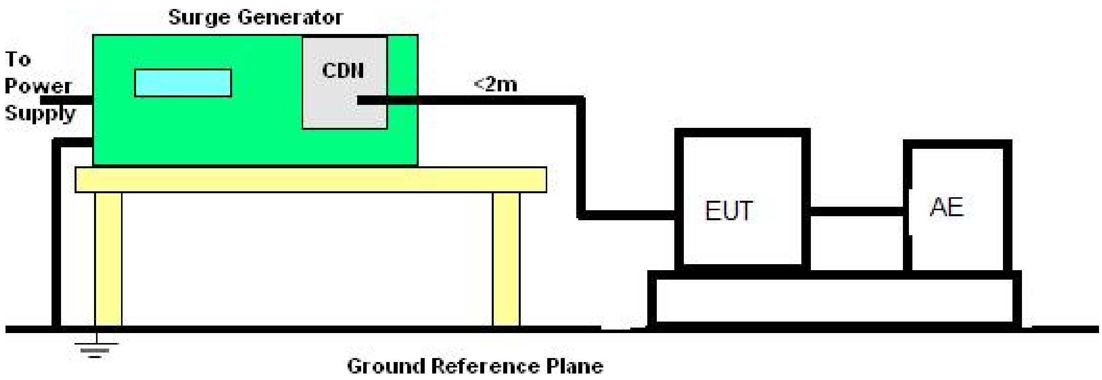
Inject Line	Voltage kV	Repetition rate kHz	Duration of Test (seconds)	Inject Method	Result
AC power port	±2	5	120	Direct	B
DC power port	--	--	--	--	--
Signal Port	--	--	--	--	--
NOTE – “--” means the item is no applicable.					

8.5 TEST CONCLUSION

PASS

9. SURGE IMMUNITY TEST

9.1 Diagram of Test Setup



9.2 Applicable Standard

IEC 61000-4-5:2014+A1:2017

AC power port: Line to line: ±1 kV, 1.2/50（8/20）

Line to earth: ±2 kV, 1.2/50（8/20）

9.3 Severity Levels and Performance Criterion

9.3.1 Severity levels

Test Level	Power Supply Coupling Mode	
	Line to Line kV	Line to Earth kV
1	NA	0.5
2	0.5	1.0
3	1.0	2.0
4	2.0	4.0
X	Special	Special

9.3.2 Performance criterion: B

9.4 Test Result

TEMPERATURE:	22℃	HUMIDITY:	43%
TEST MODEL:	Operating	Voltage supply:	440V

AC Input Power Port						
Location	Polarity		Coupling Mode	No. of Pulse	Pulse Voltage (kV)	Result
AC power ports	--	–	--	--	--	– –
	+	-	Line to Line	10	1	P
DC power ports	--	--	--	--	--	--
	--	--	--	--	--	--
NOTE "--" means the item is no applicable.						

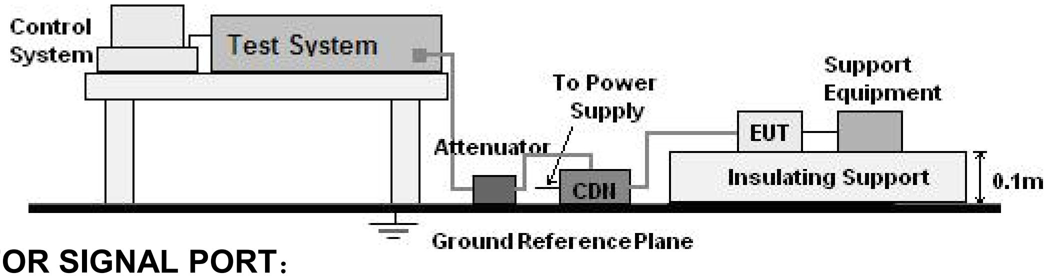
9.5 TEST CONCLUSION

PASS

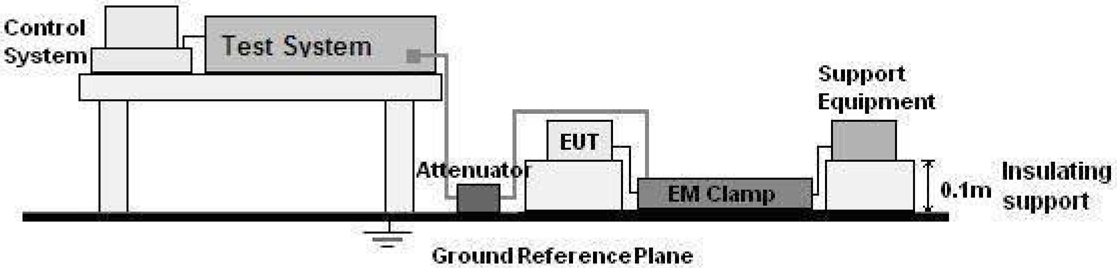
10. CONDUCTED DISTURBANCES IMMUNITY TEST

10.1 Diagram of Test Setup

FOR POWER PORT:



FOR SIGNAL PORT:



10.2 APPLICABLE STANDARD

IEC 61000-4-6:2013+Cor 1:2015, AC and DC Power ports： 0.15-80MHz, 10V Unmodulation, 80%AM (1kHz), Signal ports: 0.15-80MHz, 10V Unmodulation, 80%AM (1kHz)

10.3 SEVERITY LEVELS AND PERFORMANCE CRITERION

10.3.1 Severity levels

Frequency Range 0.15 MHz – 80 MHz		
Level	Voltage Level (e.m.f.)	
	U0 dB(μV)	U0 (V)
1.	120	1
2.	130	3
3.	140	10
Xa	Special	
Xa is an open level.		

10.3.2 Performance criterion: A

10.4 TEST RESULTS

TEMPERATURE:	22℃	HUMIDITY:	43%
TEST MODEL:	Operating	Voltage supply:	440V

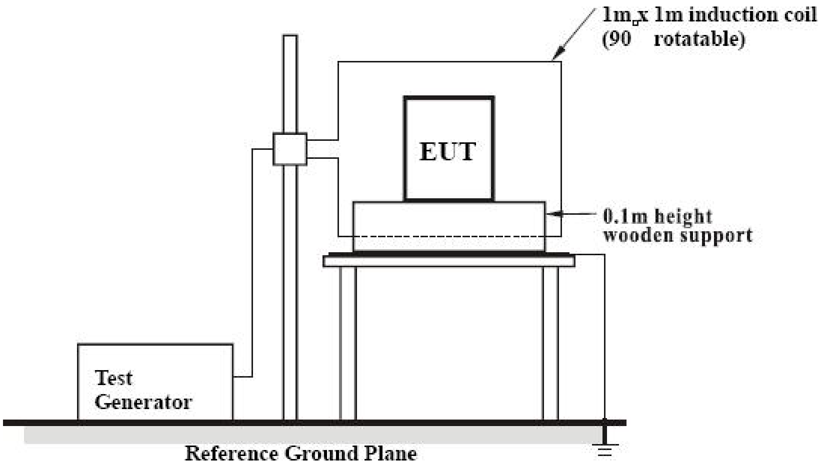
Injected Position	Frequency Range(MHz)	Strength (Unmodulated)	Results
AC power ports	0.15 ~ 80	10V(r.m.s)	A
DC power ports	---	--	--
Signal ports	---	--	--
NOTE “--” means the item is no applicable.			

10.5 TEST CONCLUSION

PASS

11. POWER-FREQUENCY MAGNETIC FIELD IMMUNITY TEST

11.1 DIAGRAM OF TEST SETUP



11.2 APPLICABLE STANDARD

IEC 61000-4-8:2009, Magnetic field strength: 10A/m, 50Hz

11.3 SEVERITY LEVELS AND PERFORMANCE CRITERION

11.3.1 Severity level:

Test Level	Magnetic field strength A/m
1	1
2	3
3	10
4	30
5	100
X	Special

11.3.2 Performance criterion: A

11.4 TEST RESULTS

TEMPERATURE:	22℃	HUMIDITY:	43%
TEST MODEL:	Operating	Voltage supply:	440V

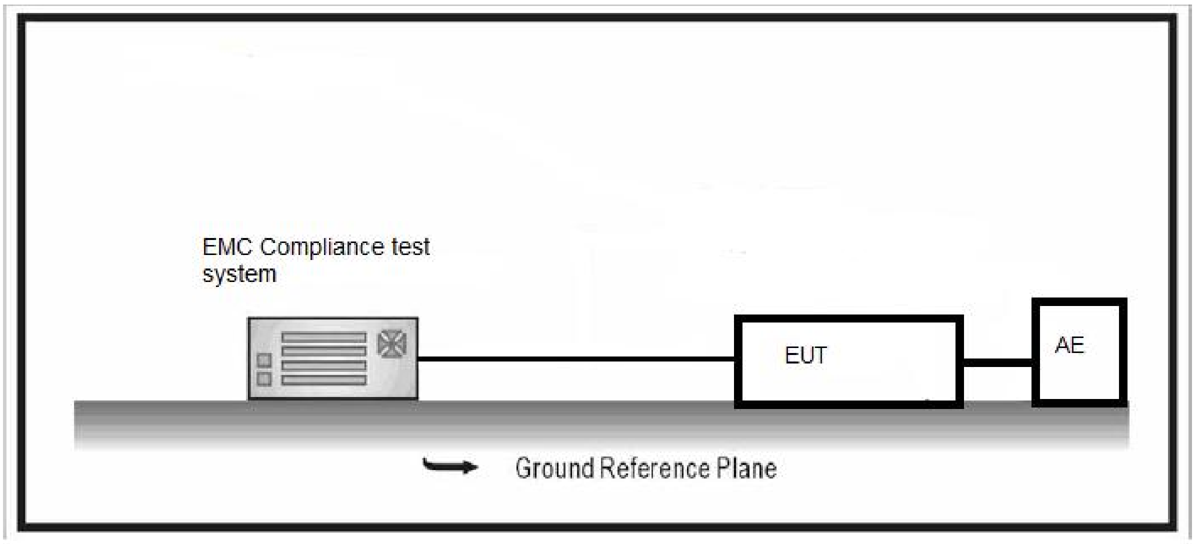
Test Level (A/m)	Testing Duration (in second)	Coil Orientation	Criterion
10	120	Axis-X	A
10	120	Axis-Y	A
10	120	Axis-Z	A

11.5 TEST CONCLUSION

PASS

12. HARMONIC CURRENT EMISSION REVIEW

12.1 DIAGRAM OF REVIEW SETUP



12.2 APPLICABLE STANDARD

EN IEC 61000-3-2:2019+A2:2024

12.3 HARMONIC CURRENT LIMITS

Limits for Class A equipment	
Harmonics Order n	Max. permissible harmonics current A
Odd harmonics	
3	2.30
5	1.14
7	0.77
9	0.40
11	0.33
13	0.21
15≤n≤39	0.15X15/n
Even harmonics	
2	1.08
4	0.43
6	0.30
8≤n≤40	0.23X8/n

12.4 REVIEW RESULTS

Temperature: 25℃ Humidity: 50% Rh Atmospheric Pressure: 1002 mbar
Test mode :Normal Working_keep EUT normal running continua

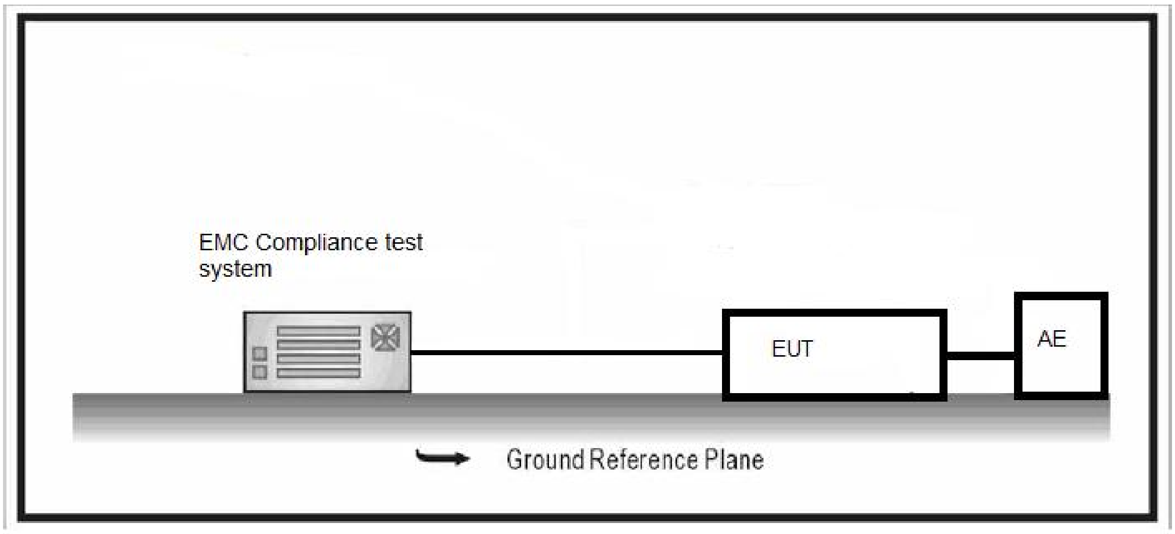
Harmonic	Status	Avg (A)	Avg L(A)	Avg %ofL	Peak (A)	Peak L(A)	Peak %ofL
1	PASS	0.36425	No Limit	N/A	0.3645	No Limit	N/A
2	PASS	0.0001	1.08	0.009559	0.00027	1.62	0.016873
3	PASS	0.34089	2.3	14.8213	0.34133	3.45	9.893623
4	PASS	0.00032	0.43	0.074881	0.00042	0.645	0.065122
5	PASS	0.30195	1.14	26.48684	0.30273	1.71	17.70351
6	PASS	0.00044	0.3	0.145803	0.00054	0.45	0.120938
7	PASS	0.25029	0.77	32.5052	0.25146	1.155	21.77143
8	PASS	0.00045	0.23	0.193478	0.00059	0.345	0.172043
9	PASS	0.19305	0.4	48.2625	0.19457	0.6	32.42833
10	PASS	0.00041	0.184	0.224772	0.00055	0.276	0.197909
11	PASS	0.1382	0.33	41.87879	0.13993	0.495	28.26869
12	PASS	0.00029	0.15333	0.191737	0.00043	0.23	0.185674
13	PASS	0.09452	0.21	45.00953	0.09627	0.315	30.56191
14	PASS	0.00012	0.13143	0.089021	0.00031	0.19715	0.155657
15	PASS	0.0698	0.15	46.53333	0.07125	0.225	31.66444
16	PASS	3.5E-05	0.115	0.03021	0.00022	0.1725	0.126864
17	PASS	0.06322	0.13235	47.76804	0.06431	0.19853	32.39391
18	PASS	5.6E-05	0.10222	0.054777	0.00025	0.15333	0.162871
19	PASS	0.0624	0.11842	52.69465	0.06344	0.17763	35.71469
20	PASS	0.00013	0.092	0.139196	0.0003	0.138	0.219123
21	PASS	0.05825	0.10714	54.36905	0.05945	0.16071	36.99459
22	PASS	0.00012	0.08364	0.138625	0.0003	0.12545	0.239753
23	PASS	0.04965	0.09783	50.75134	0.05102	0.14674	34.76785
24	PASS	3.5E-05	0.07667	0.04579	0.00027	0.115	0.231938
25	PASS	0.03949	0.09	43.88111	0.04087	0.135	30.27555
26	PASS	1.7E-05	0.07077	0.024071	0.00018	0.10615	0.167361
27	PASS	0.03199	0.08333	38.39055	0.03318	0.125	26.54171
28	PASS	9E-06	0.06571	0.01332	0.00015	0.09857	0.147153
29	PASS	0.02934	0.07759	37.81352	0.03022	0.11638	25.9686
30	PASS	9E-06	0.06133	0.014185	0.00015	0.092	0.164436
31	PASS	0.02915	0.07258	40.15652	0.02991	0.10887	27.47367
32	PASS	2.4E-05	0.0575	0.041275	0.00019	0.08625	0.215942
33	PASS	0.028	0.06818	41.07096	0.02884	0.10227	28.20099
34	PASS	2.3E-05	0.05412	0.043261	0.0002	0.08118	0.248038
35	PASS	0.02477	0.06429	38.53249	0.02573	0.09643	26.68492
36	PASS	0.00001	0.05111	0.01913	0.00011	0.07667	0.147861
37	PASS	0.02044	0.06081	33.61234	0.02141	0.09122	23.46834
38	PASS	7E-06	0.04842	0.013614	0.00011	0.07263	0.14396
39	PASS	0.01708	0.05769	29.60029	0.01785	0.08654	20.62678
40	PASS	5E-06	0.046	0.011391	8.2E-05	0.069	0.119319

12.5 REVIEW CONCLUSION

PASS

13. VOLTAGE FLUCTUATIONS AND FLICKER REVIEW

13.1 DIAGRAM OF REVIEW SETUP



13.2 APPLICABLE STANDARD

EN 61000-3-3:2013+A2:2021+AC:2022

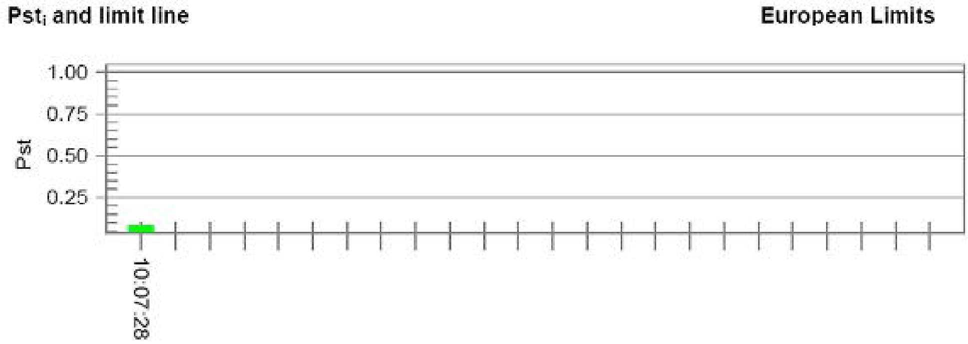
13.3 VOLTAGE FLUCTUATIONS AND FLICKER EMISSION LIMITS

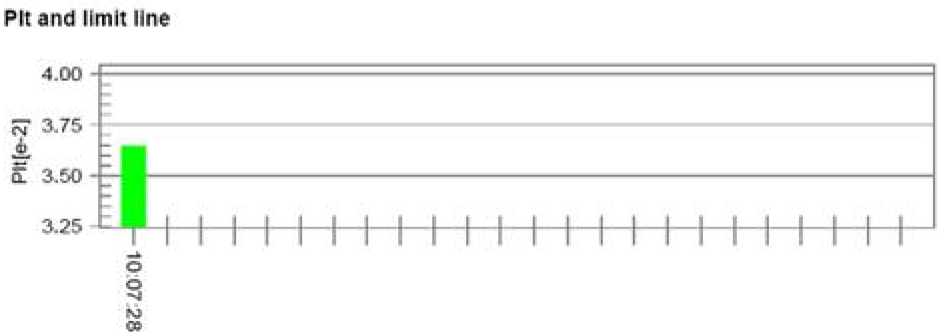
Review Item	Limit	Note
P_{st}	1.0	Short-term flicker indicator
P_{lt}	0.65	Long-term flicker indicator
T_{dt}(ms)	500	Maximum time that dt exceeds 3%
d_{max}(%)	4%	Maximum relative voltage change
d_c(%)	3.3%	Relative steady-state voltage change

13.4 REVIEW RESULTS

Temperature: 25℃ Humidity: 50% Rh Atmospheric Pressure: 1002 mbar
Test mode :Normal Working_keep EUT normal running continua

Flicker Review Summary (Run time)





Parameter values recorded during the test:

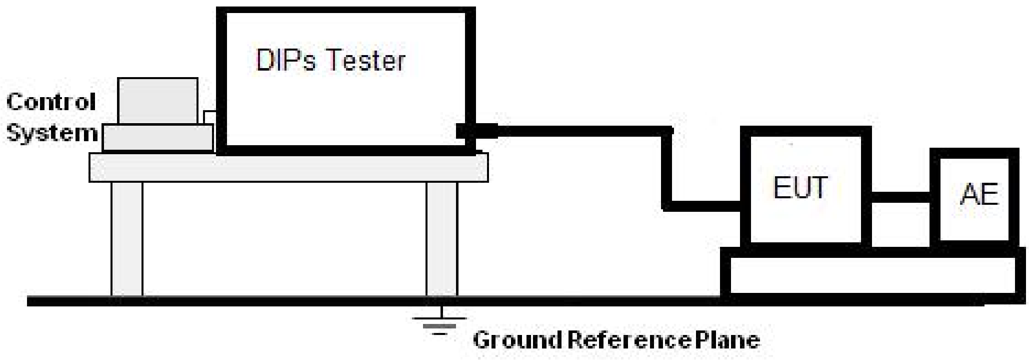
Vrms at the end of test (Volt):	229.94			
Highest dt (%):	0.62	Test limit (%):	3.30	Pass
Time(mS) > dt:	0.0	Test limit (mS):	500.0	Pass
Highest dc (%):	0.00	Test limit (%):	3.30	Pass
Highest dmax (%):	-0.59	Test limit (%):	4.00	Pass
Highest Pst (10 min. period):	0.083	Test limit:	1.000	Pass

13.5 REVIEW CONCLUSION

PASS

14. VOLTAGE DIPS

14.1 DIAGRAM OF TEST SETUP



14.2 APPLICABLE STANDARD

EN IEC 61000-4-11:2020, Test Value：Voltage dips: 0% during half cycle; 40% during 10 cycles; 70% during 25 cycles.

14.3 SEVERITY LEVELS AND PERFORMANCE CRITERION

14.3.1 Preferred severity levels and durations for voltage dips

Class[a]	Test level and durations for voltage dips (ts) (50Hz/60Hz)				
Class 1	Case-by-case according to the equipment requirements				
Class 2	0% during ½ cycle	0% during 1 cycle	70% during 25/30c cycles		
Class 3	0% during ½ cycle	0% during 1 cycle	40% during 10/12[c] cycles	70% during 25/30[c] cycles	80% during 250/300[c] cycles
Class X[b]	X	X	X	X	X

a Classes as per IEC 61000-2-4.
b To be defined by product committee. For equipment connected directly or indirectly to the public network, the levels must not be less severe than Class 2.
c “25/30 cycles” means “25 cycles for 50Hz test” and “30 cycles for 60Hz test”.

14.3.2 Preferred severity levels and durations for short interruptions:

Class[a]	Test level and durations for short interruptions (ts) (50Hz/60Hz)
Class 1	Case-by-case according to the equipment requirements
Class 2	0% during 250/300[c] cycles
Class 3	80% during 250/300[c] cycles
Class X[b]	X

a Classes as per IEC 61000-2-4.
b To be defined by product committee. For equipment connected directly or indirectly to the public network, the levels must not be less severe than Class 2.
c “250/300 cycles” means “250 cycles for 50Hz test” and “300 cycles for 60Hz test”.

14.3.3 Performance criterion:

Voltage dips: 0% during half cycle: C
40% during 10 cycles: C
70% during 25 cycles: C

14.4 TEST RESULTS

Temperature : 24℃ Humidity : 55%
Test Model : Operating Power Supply: 440V

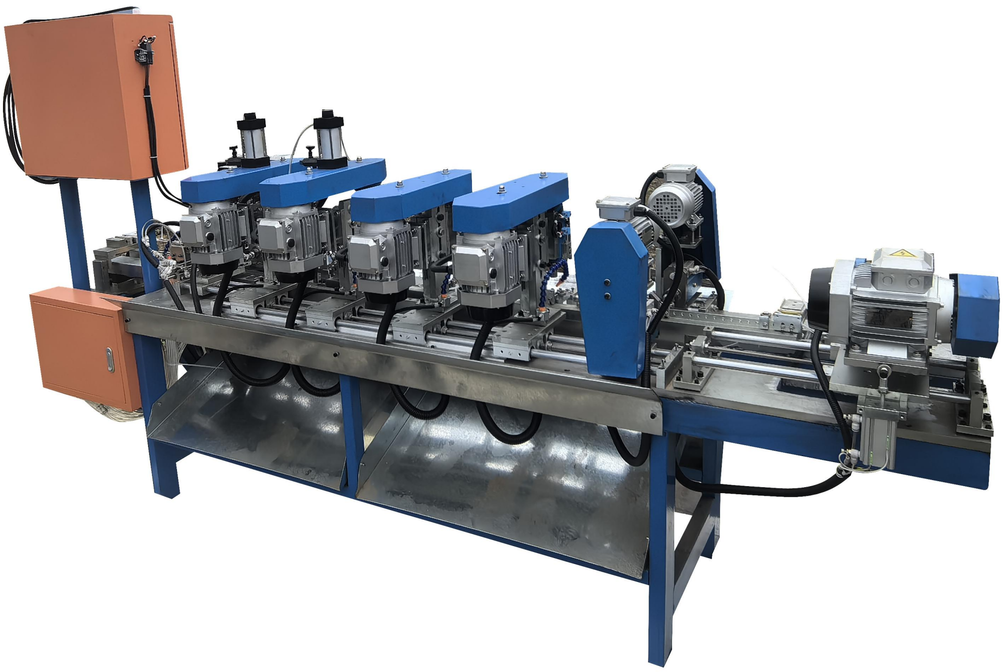
Test level ($\%U_t$)	Voltage Dips in $\%U_t$	Duration （cycles）	Phase (in angle)	Voltage phenomenon	Result
0	100	0.5	0°, 45°, 90°, 135°, 180°, 225°, 270°, 315°	Dips	P
40	60	10	0°, 45°, 90°, 135°, 180°, 225°, 270°, 315°	Dips	P
70	30	25	0°, 45°, 90°, 135°, 180°, 225°, 270°, 315°	Dips	P

14.5 TEST CONCLUSION

PASS

ANNEX 2: SAFETY PICTURES OF THE MACHINE

Type of equipment:	AUTOMATIC BRASS BUSBAR MACHINE & AUTOMATIC SCREW MACHINE

Details of:	
View: [X] general [ ] front [ ] rear [ ] right [ ] left [ ] top [ ] bottom	

Details of:	
View: [X] general [ ] front [ ] rear [ ] right [ ] left [ ] top [ ] bottom	

- End of TCF -